

# TIC 411457490

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

## BLS EPHEMERIS

Period 2.432352 d (ALIASED, see below)  
-> corrected period: 4.864705 d  
Epoch (t0) 1596.8966 BTJD  
Duration 0.1700 d  
Depth 5.531 % (55,313 ppm)  
SNR 431.2

## CATALOG CROSS-MATCH (novelty)

|                              |          |
|------------------------------|----------|
| VSX                          | no match |
| ASAS-SN                      | no match |
| Gaia eclipsing-binary table  | no match |
| Gaia variable classification | no match |
| TESS EB catalog (Prsa+2022)  | no match |
| ExoFOP TOI list              | no match |

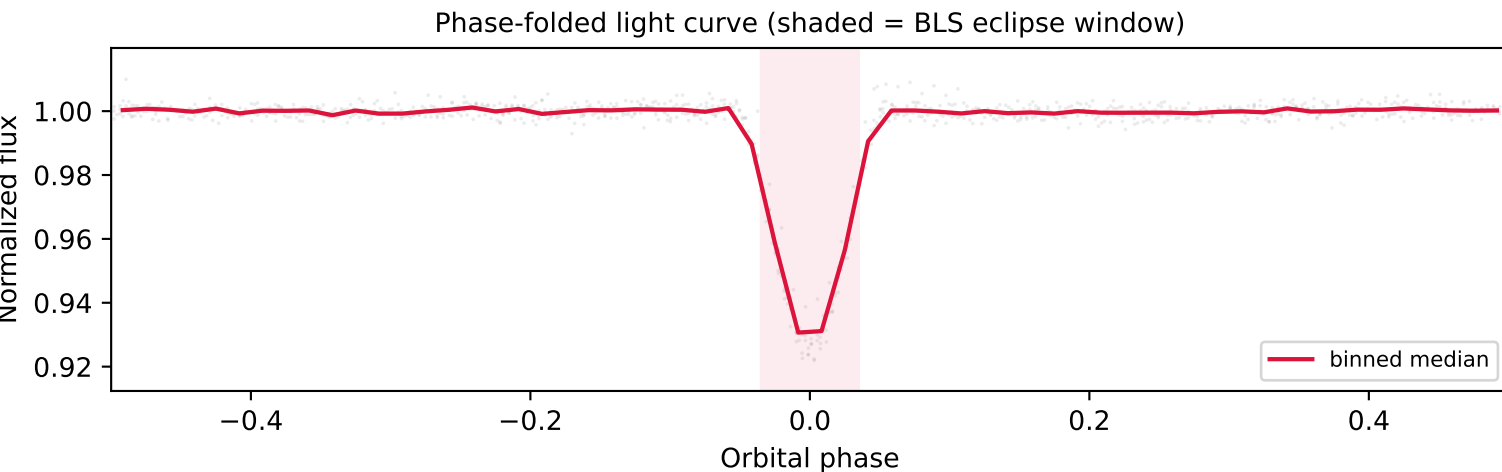
## INDEPENDENT CHECKS (f3i gates)

Eclipses observed 12 (gate  $\geq 3$ )  
Robust depth ratio 1.06 (gate  $\geq 0.50$ )  
Robust depth sigma 243.6 (gate  $\geq 7.0$ )  
Odd/even z-score 4.70 (ALIAS FLAGGED)  
Secondary eclipse not detected

Verdict: NOVEL

## PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 11 light-curve dip in this sector: 231.3 sigma t0 refined by 0.0100 phase  
Difference-image peak SNR 58.00 (gate  $\geq 5.0$ ) Compactness 0.90 (gate  $\geq 0.50$ )  
Dimming centroid offset 0.03 px (1 arcsec) (gate  $\leq 1.0$  px, i.e. on the target)  
Nearest bright-enough neighbour in cutout: TIC 411457450  
0.94 px away, dTmag +2.68 - farther from the dimming than the target



difference-image figure not found